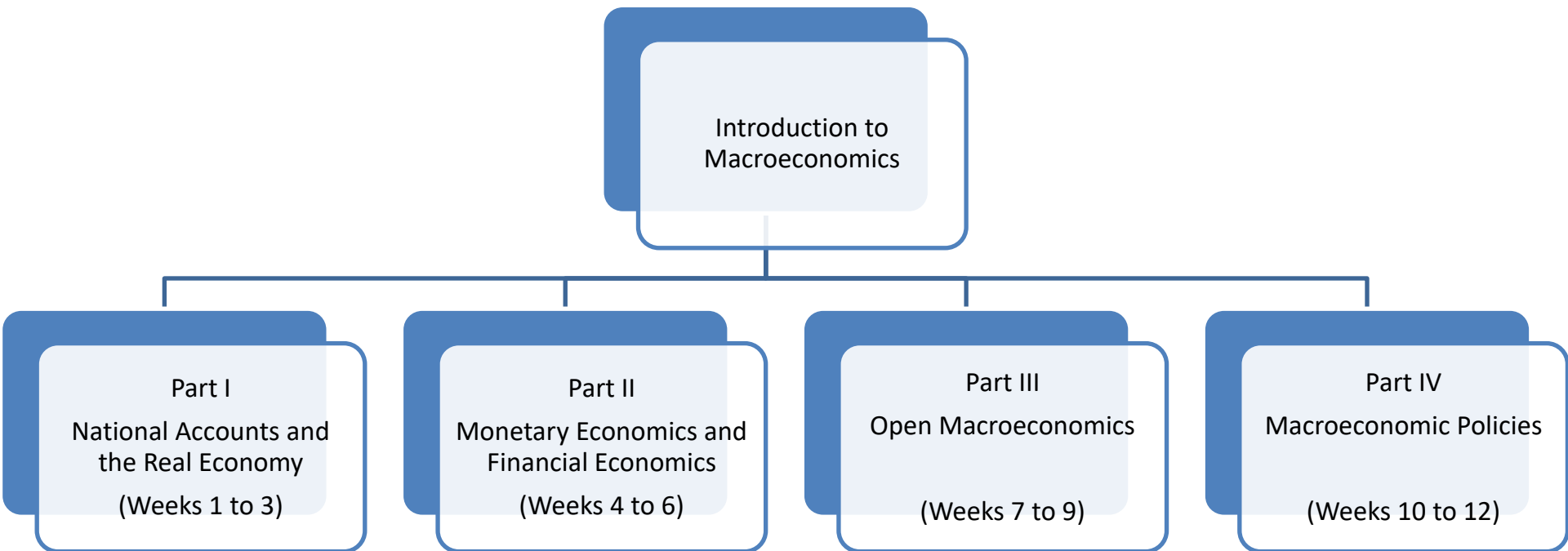


Introduction to Macroeconomics (S102016CR)

Spring Semester

<https://moodle.unige.ch/course/view.php?id=6618>

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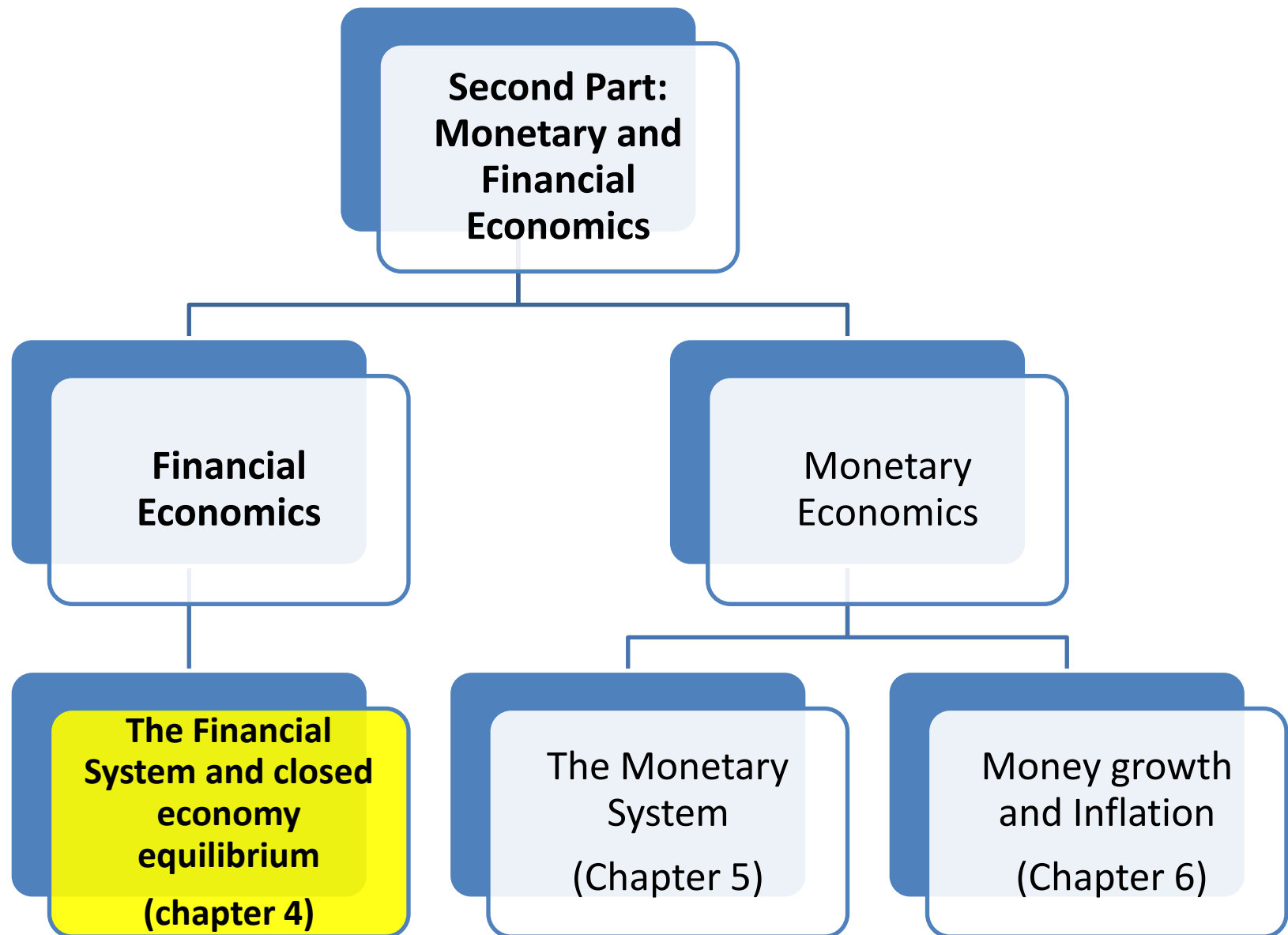
First Part:
National Accounts
and the Real
Economy

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graph TD; A["First Part:  
National Accounts  
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Economy"] --> B["Measuring a  
nation's income  
(Chapter 1)"]; A --> C["Measuring the cost  
of living  
(Chapter 2)"]; A --> D["Unemployment  
(Chapter 3)"];
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Measuring a
nation's income
(Chapter 1)

Measuring the cost
of living
(Chapter 2)

**Unemployment
(Chapter 3)**



Friday session:
For Weeks 3 and 4

Estimating the natural rate of unemployment

- Let's define the natural rate as the rate at which the number of individuals becoming unemployed equals the number of people finding jobs. Then the natural rate of unemployment is such that the total number of unemployed people remains unchanged.
- Define s as the share of those employed being separated from their jobs. So $s(L-U)$ is the flow of those becoming unemployed
- Define f as the share of those unemployed finding a job, fU is the flow of those finding a job.

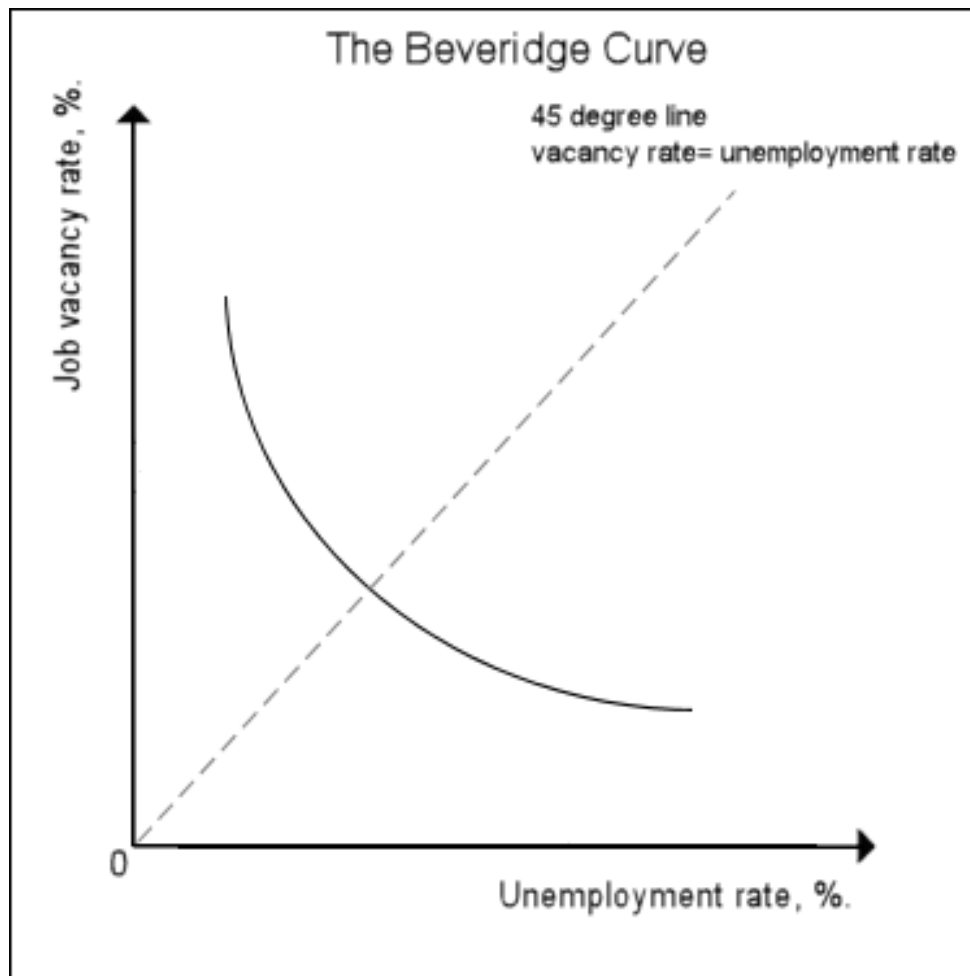
- Then the change in the number of individuals unemployed is

$$\Delta U = s(L-U) - fU$$

- Set $\Delta U = 0$, and solve for U to find the natural rate of unemployment

$$u_n = U/L = s/(s+f)$$

The **Beveridge curve** shows the relationship between job-vacancy rates and unemployment rates across time.



The intersection of the **Beveridge curve** with the 45 degree line gives us a measure of the frictional level of unemployment

When the Beveridge curve shifts outwards it signals that the labor markets are becoming less efficient. It is becoming more difficult to match unemployed workers with job vacancies.

On the other hand if the Beveridge curve shifts towards the origin it signals that the labor market is becoming more efficient.

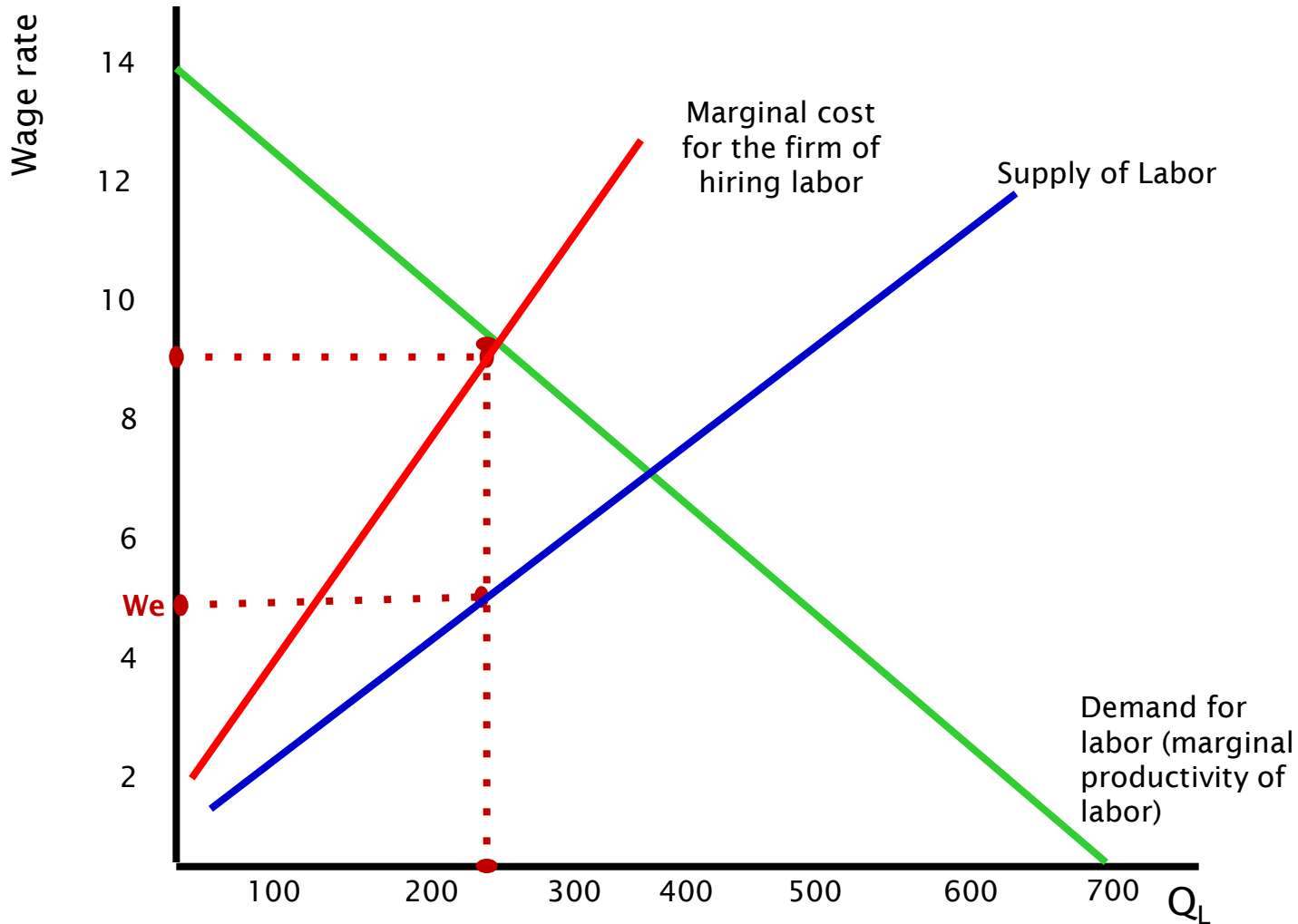
Movements along the Beveridge curve signal changes in cyclical unemployment

Short-term vs long-term unemployment:

An apparent contradiction:

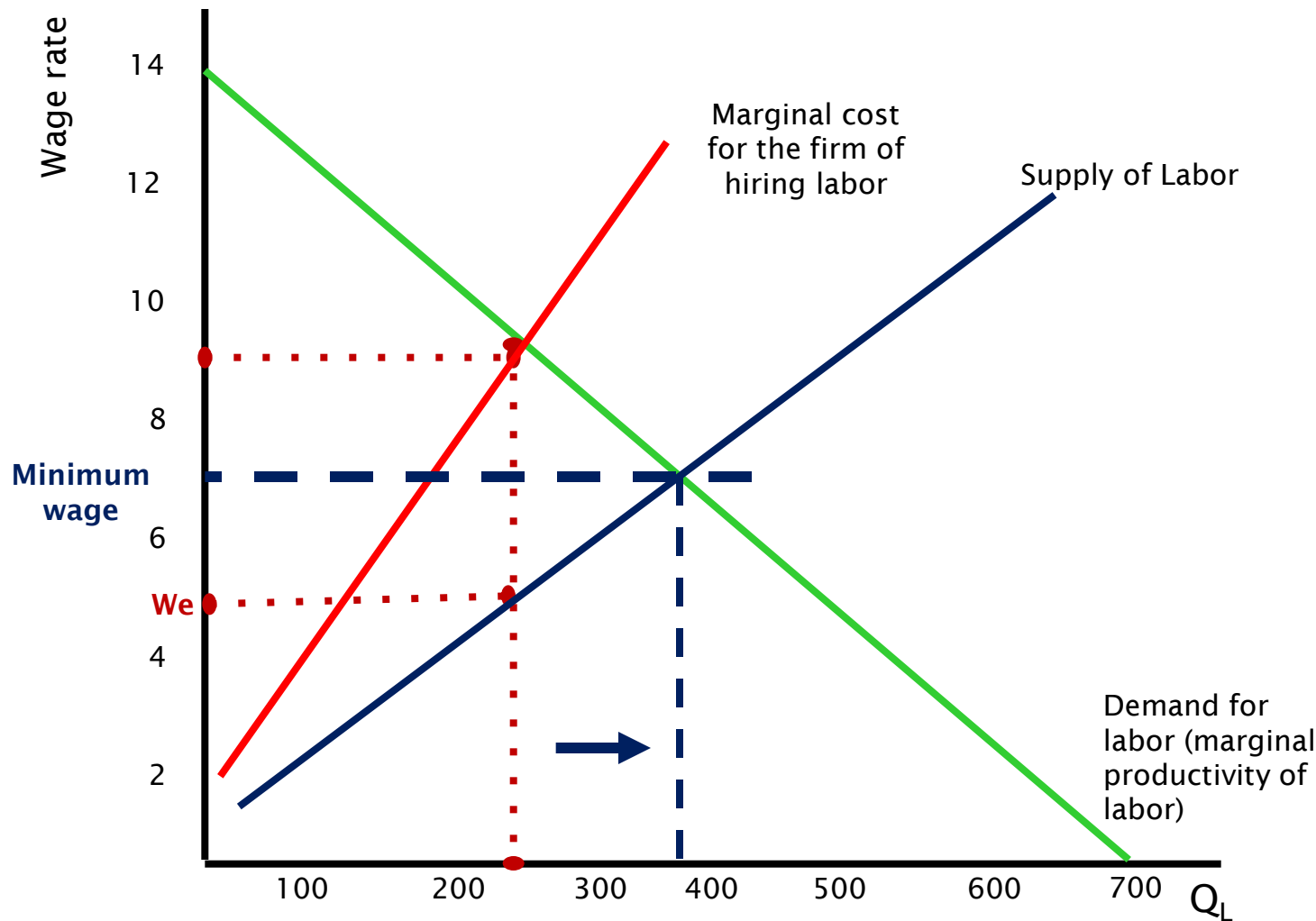
- The duration of most of unemployment is short term (if one considers the number of short term unemployed versus long term unemployed)
- But the bulk of unemployment during a certain period is long-term (if we consider the total time in short-term unemployment versus long-term unemployment in terms of hours)
- Only an apparent contradiction. Imagine an unemployment office records visits every week over one year (52 weeks). Only $\frac{1}{4}$ of those unemployed are unemployed for a week (short term). And $\frac{3}{4}$ of those unemployed are unemployed for a year (long term). But the duration of most unemployed people is short term. Indeed $\frac{1}{4} * 52 / (\frac{1}{4} * 52 + \frac{3}{4} * 52) = 94$ percent of those unemployed.

When minimum wages increase employment: The Monopsonistic Labor Market



The monopsonist sets the wage paid to workers well below their marginal productivity

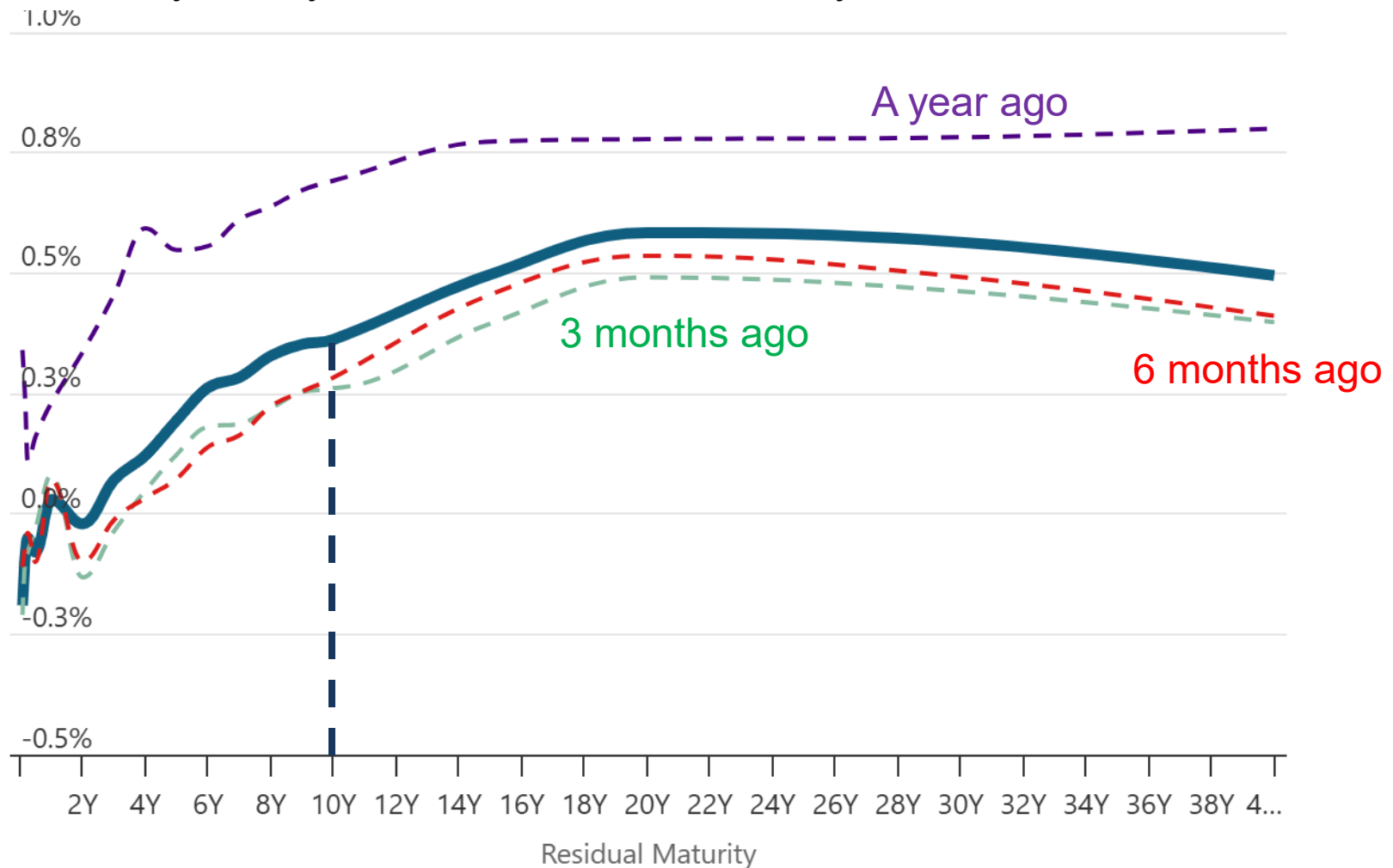
When minimum wages increase employment: The Monopsonistic Labor Market



The introduction of a minimum wage increases employment!

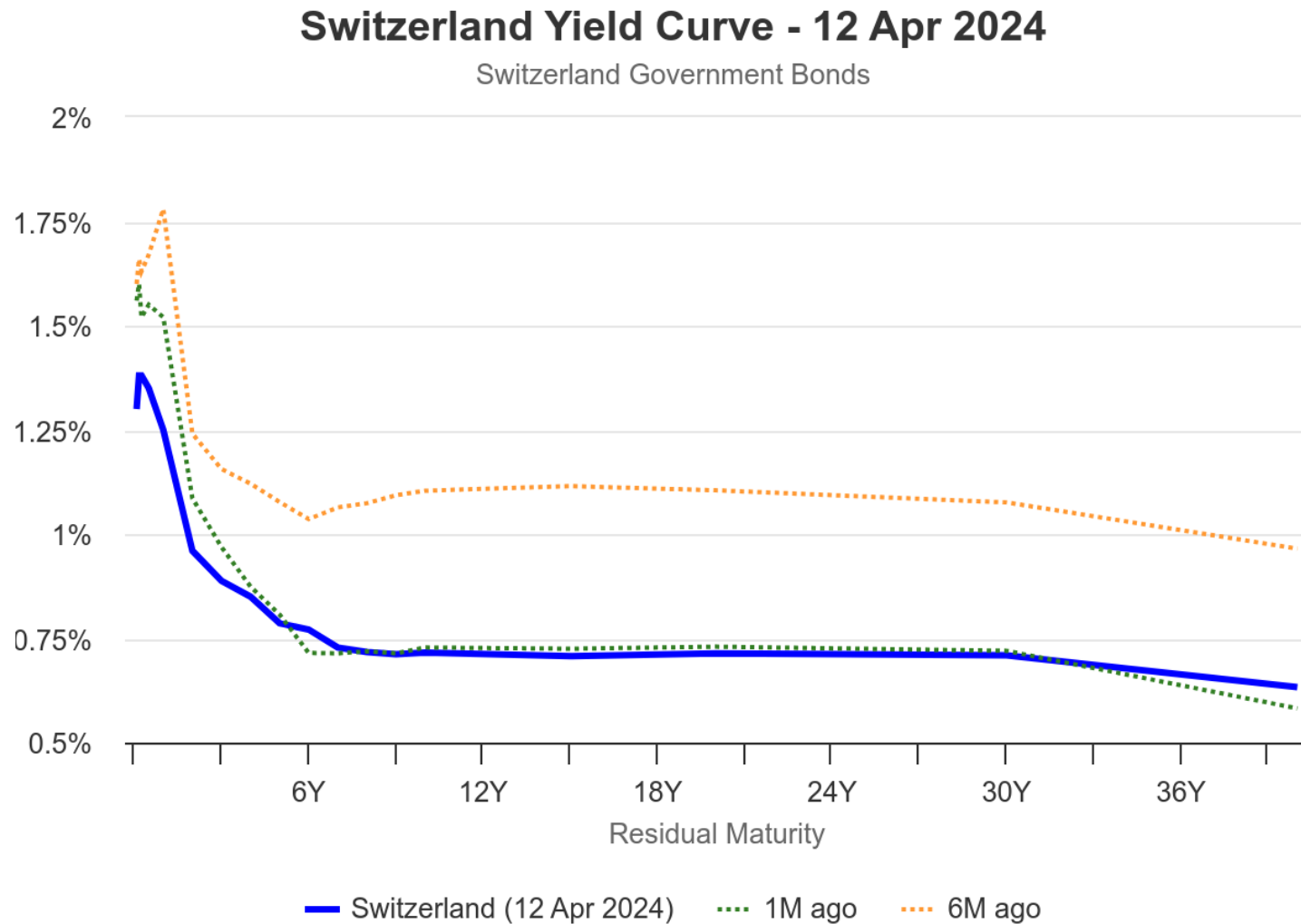
Swiss Government Bonds Yield Curve 6 March 2026

A residual maturity of 10 years in a Swiss bond had a yield of 0.38% in March 2026



Swiss Government Bonds Yield Curve in April 2024

Why was the yield curve inverted in April 2024? Why did it shift down from its level in November 2023?

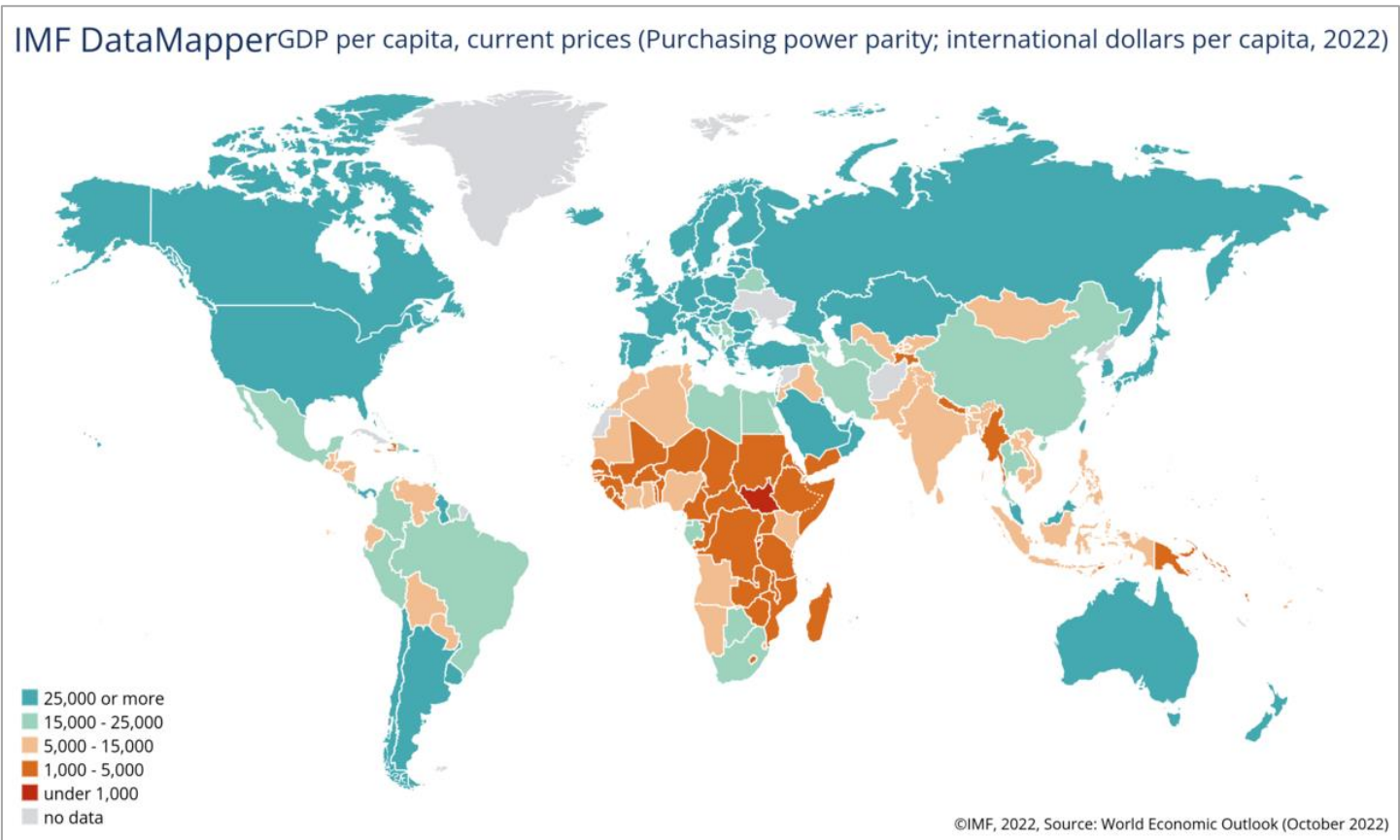


Evolution of the price of a Greek bond with a 6% nominal interest rate and a term in 2019 from March 2009 to March 2010.



How does the curve for the yield of this Greek bond look like between March 2009 and March 2010?

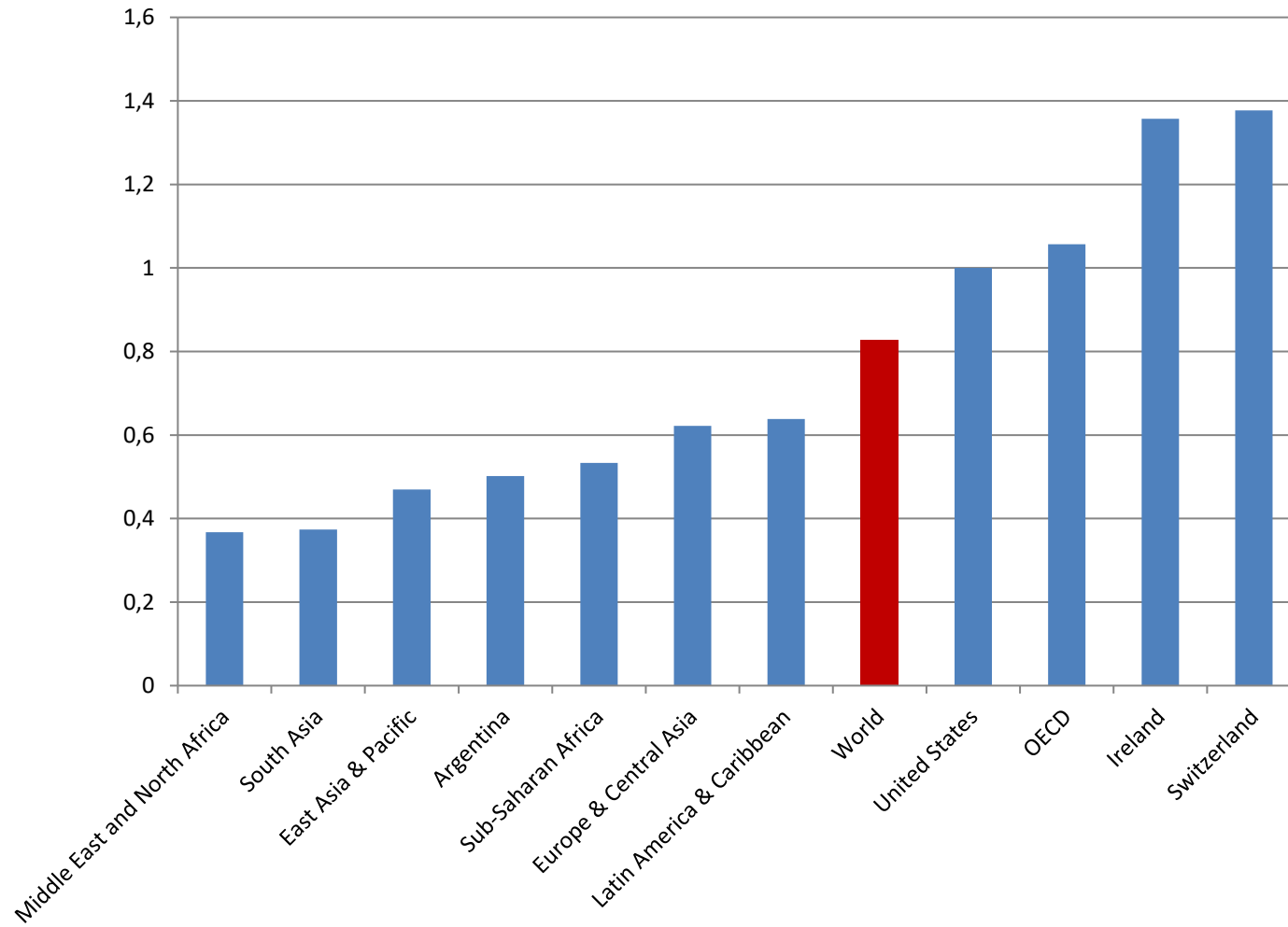
Large differences in income per capita across countries



Is GDP in Bangladesh comparable to GDP in Switzerland? Not really...

- They are not comparable because USD 1 may buy consumers much more in Bangladesh than in Japan
- Like in the case of comparisons of income across time within countries, when comparing across countries within the same period, we need to adjust for differences in prices across countries
- The adjustment factor is called the Purchasing Power Parity (PPP) index (constructed in the same way as the CPI)
- We construct a fixed consumption basket and look at differences in this consumption basket across countries, typically the US as base 100.
- If it is an index, it has no units, and therefore, the cost of the consumption basket in other countries has been transformed into USD (Otherwise, it can also be expressed in domestic currency per USD)
- The PPP suffers from the same problems and biases as the CPI

The WB and the IMF have a joint project for the construction of PPP (2025)



Understanding determinants of long-term growth in the closed economy

It is not only about increasing savings or investment. Or rather increasing output will lead to increases in both savings and investment.

A **production function** describes the relationship between inputs used in production and the quantity of output:

$$Y = A F(L, K, H, N)$$

Note that the technological parameter appears outside F , which implies that if we were to double A (our technological knowledge), we could double output.

We will also assume that the production function exhibits **constant returns to scale** in L , K , H and N . That is, a doubling of all inputs ($\alpha = 2$) causes the amount of output to double as well:

$$\alpha * Y = A F(\alpha * L, \alpha * K, \alpha * H, \alpha * N)$$

Constant returns to scale have then an interesting implication. Set $\alpha = 1/L$,

$$Y/L = A F(1, K/L, H/L, N/L)$$

It implies that labor productivity will depend on the relative quantities available of other factors of production (non-labor).

The larger the stock of physical or human capital and natural resources with respect to labor, the higher the productivity in this country (i.e., the capacity to produce goods and services per hour worked)

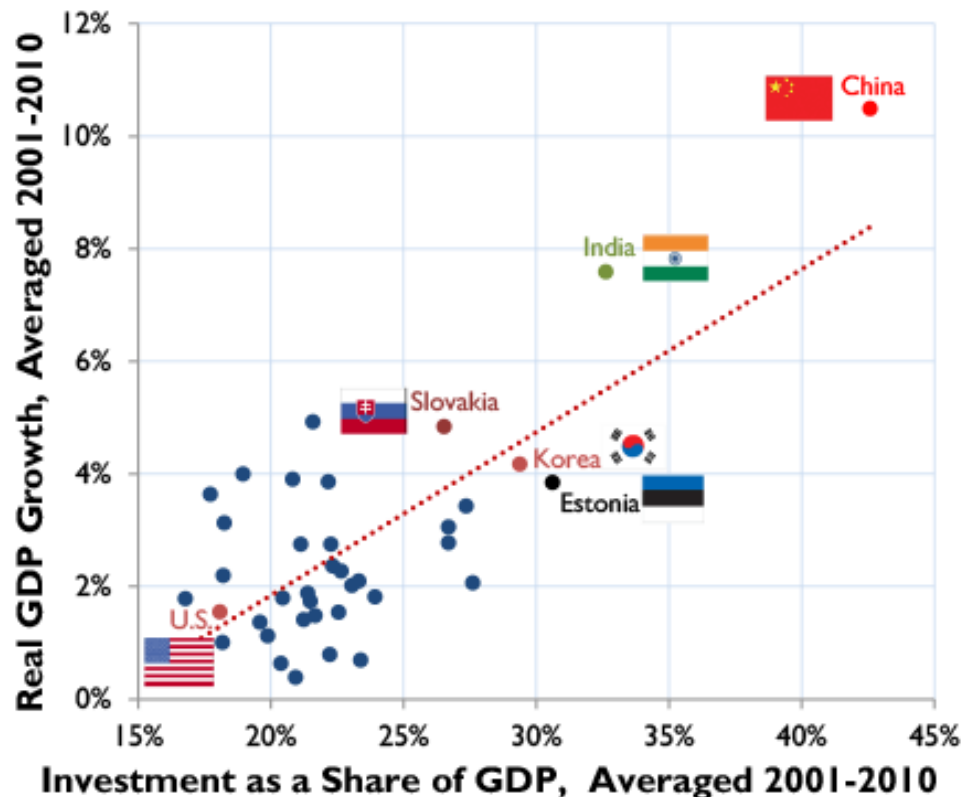
Any government policy that would help increase the quantities available of these factors of production and A will help increase productivity and lead to economic growth and higher living standards.

What is the government's role?

1. Increasing the stock of physical capital will raise future productivity.

This can be done by devoting more resources to the production of capital. How? By encouraging savings that will help finance investment (e.g., tax deductions). But also, the encouragement of foreign investment. Foreign investment can take two forms: FDI and portfolio investment.

Figure 6. Investment Drives Long-run Economic Growth
Investment Versus Real GDP Growth in OECD and BRIC Countries



2. An increase in human capital will also help increase productivity.

For a country's long-run growth, education is at least as important as investment in physical capital.

- In the developed economies of Western Europe and North America States, each year of schooling raises a person's average wage by about 10 percent.
- But there are also positive externalities from a person's education. An educated person might generate new ideas about how best to produce goods and services, which might enter society's pool of knowledge and provide an external benefit to others.

Thus, one way the government can enhance the standard of living is to provide schools and encourage the population to take advantage of them.

In developing countries, governments recognizing the importance of education for long run growth (and the cost of education for families in the short run) have introduced “Conditional Cash Transfer” programs ([Progreso in Mexico](#)).

3. A good administration of Natural Resources is important for sustainable growth.

Some talk about the «curse of natural resources». But the problem is not the stock of natural resources, but corruption and the lack of good institutions, which lead to a strong dependence on the stock of natural resources.

Norway and Qatar, the third and fifth most prosperous countries in the world in terms of GDP per capita, are also well endowed in natural resources.

At the same time, natural resources are optional to be highly productive. Japan is relatively poor in natural resources but is among the wealthiest countries in the world.

It imports most of the natural resources it needs from the rest of the world. So the management of natural resources (in Japan) is not crucial for Japan's productivity. Its management in the rest of the world may be, though (think CO2 emissions and the need for a border carbon tax?)

4. Encouraging Technological Progress and Knowledge will also help increase productivity.

This can be done through R&D subsidies, in particular in basic research without direct business applications

If research has a business application, it will probably be privately financed, at least in countries with good property right protection for inventors (patent system). Yet be aware of the trade-off that patents create: incentives vs. monopolies.

Technological progress increases productivity, but it there is evidence that it also destroy jobs. The increasing use of robots and AI is an example of this (ATM replacing bank tellers, self-driving cars replacing taxi drivers, etc)

But evidence that it creates jobs elsewhere: creative destruction.

Careful, however, about the consequences for inequality and sustainability of the current rapid trend. Great Ted talk by Andrew McAfee (MIT):

https://www.youtube.com/watch?v=cXQrbxD9_Ng

5. Other government policies with positive effects on growth:

- The rule of law, which strengthens capital investment and technological progress
- Macroeconomic stability, which promotes investment and business initiatives;

The Exam

Material covered

- All that has been seen in class or seminar
- The companion chapters in M&T or K&W if the material was covered in class or seminar
- No surprises in terms of material covered
- No differences between the English and French exams (apart from language).

Type of exam

- 2 hours
- 30 MCQ with 4 possible answers. No penalties → Answer all questions
- There will be 6 to 8 questions on the first part of the course, 6 to 8 questions on the second part, 6 to 8 questions on the third part, and 6 to 8 questions on the fourth part.
- Over the entire exam
 - 1/3 of the questions will require some simple calculus
 - 2/3 of the questions will be analytical
- A two-sided A4 page of personal notes and a non-programmable calculator are allowed

Examples of exam questions for weeks 3 and 4

votamatic.unige.ch

code d'accès: PHZK



<https://votamatic.unige.ch/form/ErshQibpm7XcheFJtYSI>